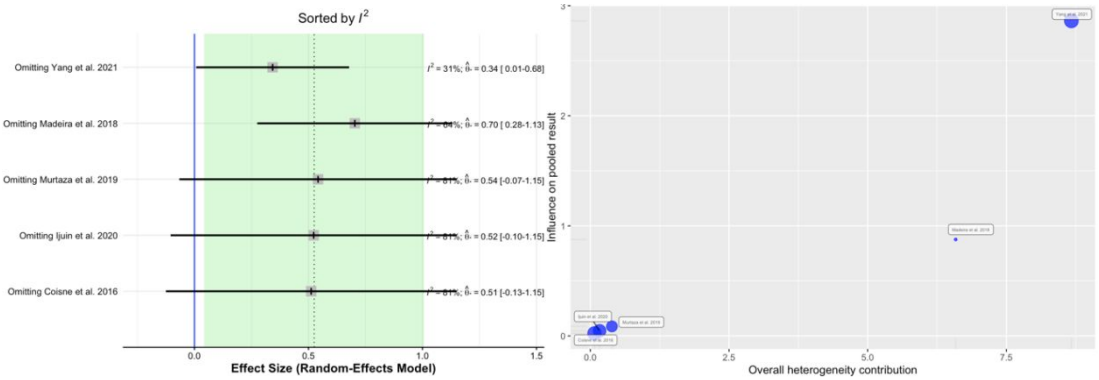
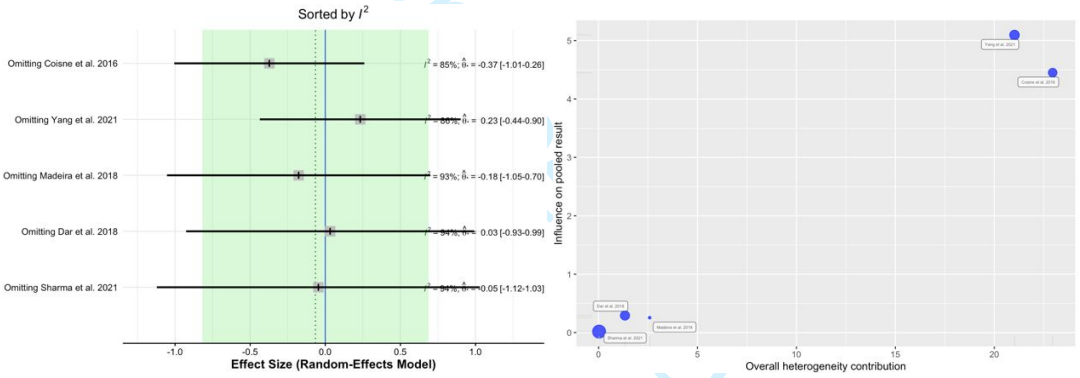
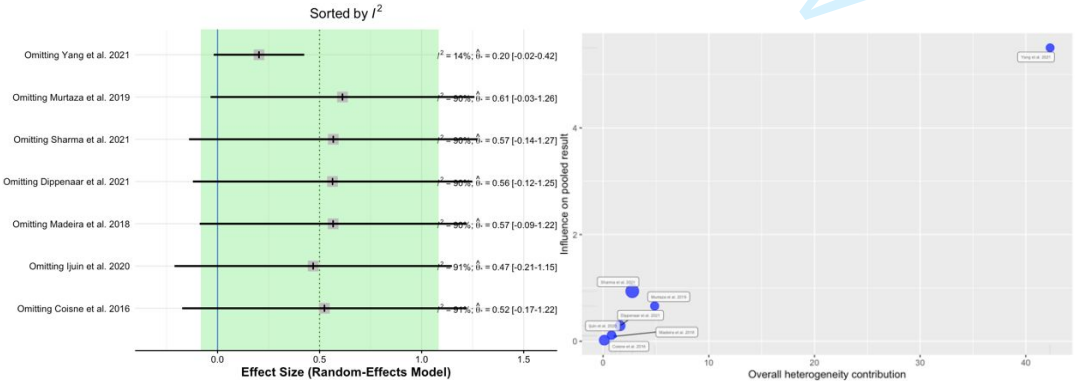
Supplementary Figure 3-A: Leave-one-out analysis and baujat plot of TAEF



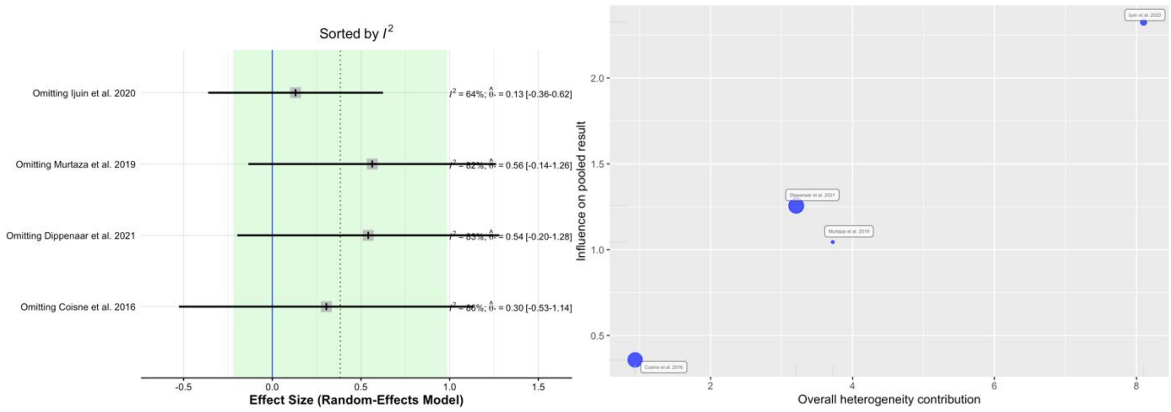
Supplementary Figure 3-B: Leave-one-out analysis and baujat plot of LA volume



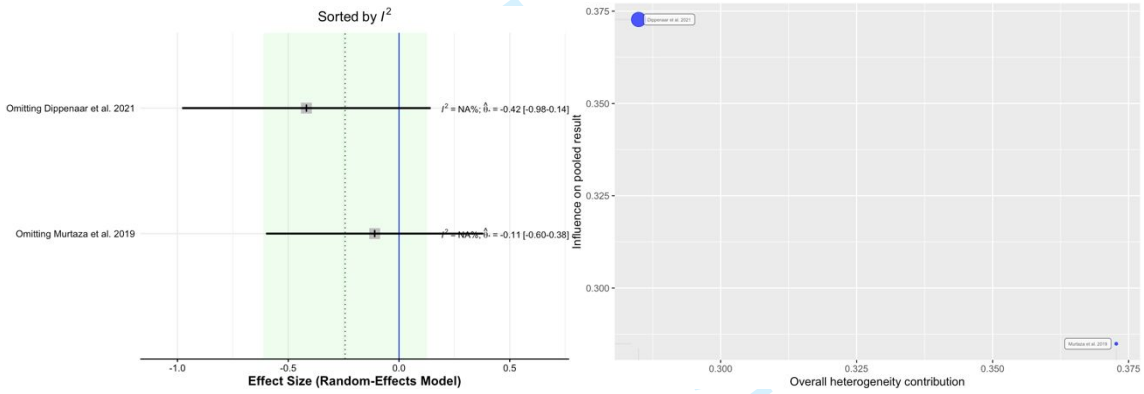
Supplementary Figure 3-C: Leave-one-out analysis and baujat plot of PALS



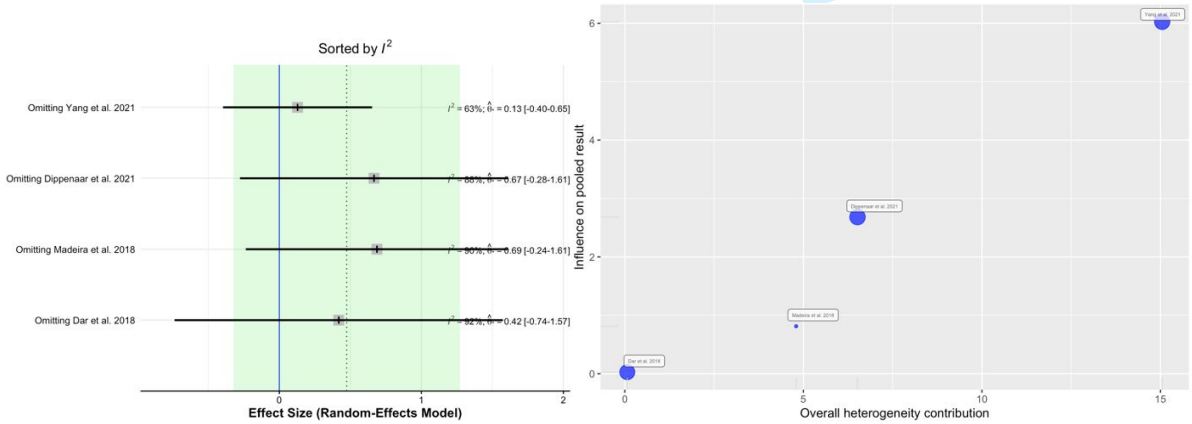
Supplementary Figure 3-D: Leave-one-out analysis and baujat plot of PACS



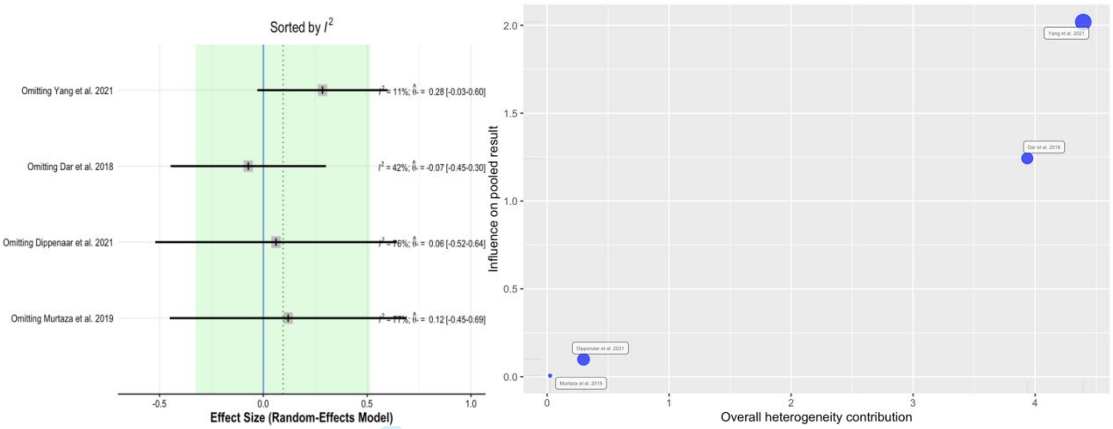
Supplementary Figure 3-E: Leave-one-out analysis and baujat plot of the strain during atrial contraction



Supplementary Figure 3-F: Leave-one-out analysis and baujat plot of the strain during ventricular systole



Supplementary Figure 3-G: Leave-one-out analysis and baujat plot of the strain during ventricular diastole



Supplementary table 1:

Newcastle-Ottawa scale for Quality assessment of the included studies

	Selection				Comparability		Outcome			
Study	Representativeness of exposed cohort	Selection of non- exposed cohort	Ascertainment of exposure	Demonstration that outcome of interest was not present at start of study	Adjust for the most important risk factors	Adjust for other risk factors	Assessment of outcome	Follow-up length	Loss to follow-up rate	**Total quality score**
Marta Madeira et al/2018	0	0	1	1	0	0	1	1	1	**5**
Yang et al /2021	0	0	1	1	1	1	1	1	1	**7**
Shun Ijuin et al /2020	0	0	1	1	0	0	1	1	1	**5**
Augustin Coisne et al 2016	0	0	1	1	0	0	1	1	1	**5**
Ghulam Murtaza et al/2019	0	0	1	0	0	0	1	0	1	**3**
Esseim Sharma et al/2021	0	0	1	1	0	0	1	0	1	**4**
Dar/2018	0	0	1	1	0	0	1	1	1	**5**
Dippenaar/2021	NA	NA	NA	NA	NA	NA	NA	NA	NA	**NA**

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Tables

Table 1: summary of the included studies

Study	Year/Country	Study design	Sample size	Follow up	Summary of methodology	Outcome
Marta Madeira et al	2018/Portugal	Retrospective	16	N.A	TEE performed before and after LAAC. LA volumes were calculated using the biplane method, and LA mechanics were assessed using STE.. The analysis focused on the LA reservoir phase strain and strain rate.	percutaneous LAA closure does not affect LA reservoir function
Yang et al	2021/China	Retrospective	65	12 months	Intervention group: Combined AF ablation and LAAC. Control group: AF ablation. echocardiography and speckle tracking echocardiography were performed to assess LA reservoir, conduit, and contractile function	Both the combined therapy group and the simple ablation group demonstrated significant improvement in LA function. But most of the effects appeared to result from ablation, not LAAC
Shun Ijuin et al	2020/Germany	Retrospective	95	180 days	95 patients who underwent percutaneous LAAC. LA strain was evaluated at three different time intervals by TEE (baseline, 45 days, and 180 days after the procedure	The study showed improvement in TEE-derived LA strain following LAAC within 45 days of implantation.
Augustin Coisne et al	2016/France	Prospective	33	45 days	33 patients evaluated by TEE at time of discharge and 45 days.	LAA closure was associated with an improvement in LA mechanical function
Ghulam Murtaza et al	2019/USA	Prospective	25	N.A	25 patients underwent LAAC. LA function parameters	LAAC leads to improvement in all volumetric indices of the

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					(volumetric, strain indices) were assessed by speckle tracking before and after	LA. However, there was discrepancy between volumetric and strain indices
Esseim Sharma et al	2021/USA	Retrospective	67	N.A	TTE performed before and after LAAC.	LAAC resulted in elevated LV filing pressure, improvement in PALS, RVGLS, and increased LVEF
Dar et al	2018/USA	Prospective	66	N.A	TEE was performed before and after the LAAC.	LAA exclusion appears to improve the mechanical function of LA when assessed by STE.
Dippenaar et al	2021/South africa	Prospective	32	N.A	32 patients underwent LAAC. LA reservoir, conduit and contractile strain and strain rate were assessed with two-dimensional speckle tracking echocardiography	No statistically significant improvement in LA mechanical function was seen after LAAO with the Amplatz or Amulet device, although a trend towards improved strain was observed for reservoir and conduit strain, with a worsening in contractile strain.

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TABLE 2: baseline characteristics

STUDY	Age	Male	HTN	DM	CAD	CHF	Afib	CHA2DS2-VASc score	HAS-BLED score
Marta Madeira et al	71 ± 9	63%	N.A	N.A	N.A	N.A	Permanent (75%) Persistent (6%) Paroxysmal (19%)	5 [4 - 5]	3 [2 - 3]

Yang et al	61.8 ± 7.9	66%	69%	14%	23%	N.A	Persistent AF: 46% Long-standing persistent AF: 54%	3 (2, 4)	3 (2, 3)
Shun Ijuin et al	75±68	67%	97%	40%	40%	85%	Permanent: 65% Paroxysmal: 34%	4.4 ± 1.4	4.1± 0.9
Augustin Coisne et al	67.1 ± 12.2	51.5%	82%	33%	N.A	N.A	N.A	4.5 ± 1.37	3.4 ± 1
Ghulam Murtaza et al	76 ± 6.9	60%	96%	32%	76%	36%	Persistent: 56% Paroxysmal: 44%	5.0 ± 1.7	4.0 ± 1.5
Esseim Sharma et al	73.2±9.0	70%	90%	48%	68%	65.7%	N.A	4.5 ± 1.3	N.A
Dar et al	70 ± 9.23	66%	82%	30%	48%	24%	Paroxysmal: 44% Persistent: 32% Longstanding: 24%	3.7 ± 1.7	3.3 ± 1.4
Dippenaar et al	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A

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Augustin Coisne et al/2016	0	0	1	1	0	0	1	1	1	**5**
Ghulam Murtaza et al/2019	0	0	1	0	0	0	1	0	1	**3**
Esseim Sharma et al/2021	0	0	1	1	0	0	1	0	1	**4**
Dar/2018	0	0	1	1	0	0	1	1	1	**5**
Dippenaar/2021	NA	NA	NA	NA	NA	NA	NA	NA	NA	**NA**

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